Q1

Directions for question 1: Type in your answer in the input box provided in the question.

P sold an article to Q at 25% profit. Q sold the article to R at 20% profit. If R's cost price is Rs.200 more than P's cost price, the cost price for P is Rs.

Solution:

Let the cost price of P be Rs.x.

P sold the article to Q at 25% profit.

$$\text{P's profit} = \frac{25}{100}x = \frac{x}{4}.$$

$$\text{P's selling price} = x + \frac{x}{4} = \frac{5x}{4}$$

This is Q's cost price.

$$\text{Q's profit} = \frac{20}{100}\left(\frac{5x}{4}\right) = \frac{x}{4}$$

$$\text{Q's selling price} = \frac{5x}{4} + \frac{x}{4} = \frac{3x}{2}$$

This is also R's cost price

$$\frac{3x}{2} - x = 200$$

$x = 400.$

Ans : (400)

Directions for questions 2 to 4: Select the correct alternative from the given choices.

If f(x, y, z) = 2x + 3y + 4z where x, y and z are numbers in the decimal system, then $f((315)_7, (123)_5, (648)_9)$ = _____ .

- a) $(9F8)_{16}$
- b) $(89F)_{16}$
- c) $(8F9)_{16}$
- d) $(F89)_{16}$

Solution:

$(315)_7 = 3 \times 7^2 + 1 \times 7 + 5 \times 7^0$
$= 147 + 7 + 5 = (159)_{10}$
$(123)_5 = 1 \times 5^2 + 2 \times 5^1 + 3 \times 5^0 = 25 + 10 + 3 = (38)_{10}$
$(648)_9 = 6 \times 9^2 + 4 \times 9 + 8 \times 9^0 = 486 + 36 + 8 = (530)_{10}$
$\therefore f((315)_7, (123)_5, (648)_9)$
$= f(159, 38, 530) = 2(159) + 3(38) + 4(530)$
$= 318 + 114 + 2120 = (2552)_{10}$

$16 \mid 2552$
$16 \mid 159 - 8$
$9 - 15$

$\therefore (2552)_{(10)} = (9F8)_{16}$ Choice (A)

Directions for questions 2 to 4: Select the correct alternative from the given choices.

Find the ratio of the 4 parts into which a certain sum is to be divided such that when the first, second, third and fourth parts are invested for 2 years, 3 years, 4 years and 5 years respectively at the same rate of simple interest, equal interests are earned.

- a) 30 : 20 : 15 : 12
- b) 12 : 15 : 20 : 30
- c) 2 : 3 : 4 : 5
- d) 5 : 4 : 3 : 2

Solution:

Let the first, second, third and fourth parts be ₹P_1, ₹P_2, ₹P_3 and ₹P_4 respectively.
Let the rate of interest at which each part is lent be R% p.a.

$$\frac{P_1 R(2)}{100} = \frac{P_2 R(3)}{100} = \frac{P_3 R(4)}{100} = \frac{P_4 R(5)}{100} = k \text{ (say)}$$

$$\therefore P_1 : P_2 : P_3 : P_4 = \frac{100}{2R} : \frac{100}{3R} : \frac{100}{4R} : \frac{100}{5R}$$

$= 30 : 20 : 15 : 12$

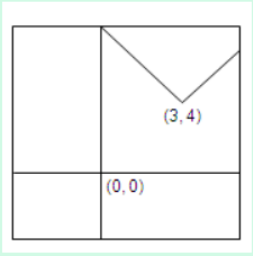
Choice (A)

Directions for questions 2 to 4: Select the correct alternative from the given choices.

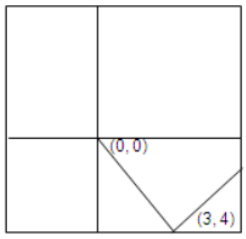
Which of the following graphs represents $y =$

$|x-3|+4$

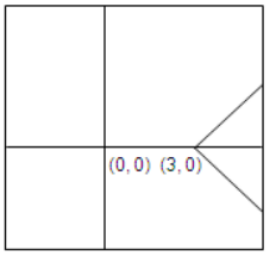
?

a) 

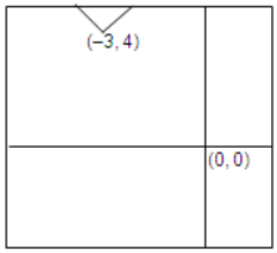
(Your answer - ☑)

b) 

c)



d)



***Directions for question 5**:* Type in your answer in the input box provided below the question.

If a = $\sqrt{\frac{9+4\sqrt{5}}{9-4\sqrt{5}}}$,

find the value of $a^2(a-18)^2$.

Solution:

$$a = \sqrt{\frac{9+4\sqrt{5}}{9-4\sqrt{5}}} \times \sqrt{\frac{9+4\sqrt{5}}{9+4\sqrt{5}}}$$

$$= \frac{9+4\sqrt{5}}{\sqrt{81-80}} = 9+4\sqrt{5}$$

$$\therefore a^2(a-18)^2$$

$$= \left(9+4\sqrt{5}\right)^2\left(9+4\sqrt{5}-18\right)^2 = \left[\left(4\sqrt{5}+9\right)\left(4\sqrt{5}-9\right)\right]^2$$

$$= (80-81)^2 = 1$$

Ans : (1)

Directions for question 6: Select the correct alternative from the given choices.

A man divided his property among his two sons and three daughters in the ratio 2.3 : 2.5 : 3.2 : 4.7 : 2.6 respectively. The 3rd daughter received Rs.5200. What is the difference in the amounts (in Rs.) received by the daughter who received the maximum amount and the son who received the minimum amount?

- a) 2000
- b) 3200
- c) 4800
- d) 12000

Solution:

Let the share of the youngest daughter be ₹2.6x.
2.6x = 5200
⇒ x = 2000
Amounts received by the daughter who received the maximum amount = ₹4.7x.
Amount received by the son who received the minimum amount = ₹2.3x.
Required difference = ₹ (4.7x – 2.3x)
= ₹2.4x = Rs.4800.
Choice (C)

Directions for question 7: Type in your answer in the input box provided in the question.

The difference in compound interest accrued on Rs.20,000 for one year at 6% p.a. when interest is compounded annually and when it is compounded half-yearly is Rs.

Solution:

When compounded annually,

$$\text{interest } I_1 = \frac{P \times R}{100} = \frac{20000 \times 6}{100} = 1200$$

When compounded half-yearly interest

$$I_2 = \frac{P}{100} \times \frac{r}{2} + \left(P + \frac{P}{100}\frac{r}{2}\right) \times \frac{r}{200}$$

$$I_2 = \frac{20000 \times 3}{100} + \left(20000 + \frac{20000 \times 3}{100}\right) \times \frac{3}{100}$$

$$= 1218$$

Difference = $I_2 - I_1$ = 1218 – 1200 = ₹18

Ans : (18)

Directions for question 8: Select the correct alternative from the given choices.

If $x^2 > x^3$ and $x^4 > x^2$, which of the following can be concluded about x?

- a) x lies between 0 and 1
- b) x lies between −1 and 0
- c) x lies between $-\infty$ and −1
- d) x lies between 1 and ∞

Solution:

$x^2 > x^3 \Rightarrow x^2(1 - x) > 0$
As $x^2 > 0$, $1 - x > 0$
$\therefore 0 < |x| < 1$ or $-\infty < x < -1$
$x^4 > x^2$
$\Rightarrow x^2(x^2 - 1) > 0.$
As $x^2 > 0$, $x^2 - 1 > 0$
$\therefore x^2 > 1.$
$\therefore 1 < x < \infty$ or $-\infty < x < -1$
As $x^2 > x^3$ and $x^4 > x^2$, $-\infty < x < -1$. Choice (C)

Directions for question 9: Type in your answer in the input box provided below the question.

In a convex nonagon, find the maximum number of points of intersection of the diagonals which lie inside the nonagon.

Solution:

Consider any 4 vertices of the convex polygon, say A, B, C, D in that order along the perimeter. A can be joined to B, C or D. In each case, the other two vertices can be joined. Thus we would get 3 pairs of lines. If we extend these lines until they intersect, we would get 3 points of intersection. We can see that while two of these points lie outside the quadrilateral ABCD (they would also lie outside the nonagon as the nonagon is convex), the third one – viz the point where AC and BD intersect would lie inside the quadrilateral ABCD (and hence inside the nonagon as it is convex). Thus, corresponding to every set of 4 vertices, there is exactly one point of the required kind. The number of such points is ${}^9C_4 = 126$. Ans : (126)

Directions for question 10: Select the correct alternative from the given choices.

A function f : N → N is defined as f(1) = 0, f(2) = 1, f(3) = 0, f(4) = 1 and so on.

Which of the following relations best describes the function f(n)?

- a) $(-1)^n - (1)^n$
- b) $(1)^n - (-1)^n$
- c) $(1)^n + (-1)^n$
- d) None of these

Solution:

We solve the problem by the elimination of choices method :

Choice (A) : $f(1) = (-1)^1 - (1)^1 = -2 \neq 0$

Choice (B) : $f(1) = (1)^1 - (-1)^1 = 2 \neq 0$

Choice (C) : $f(1) = (1)^1 + (-1)^1 = 0$ and

$f(2) = (1)^2 + (-1)^2 = 1 + 1 = 2 \neq 1$

Hence, choice (D) is the right option.

Choice (D)

Directions for question 11: Type in your answer in the input box provided in the question.

The HCF of two natural numbers is 33 and their sum is 528. The number of pairs of numbers satisfying the above conditions is .

Solution:

Let the two numbers be 33a and 33b where a and b are co-prime and a < b.

$\therefore 33(a + b) = 528$

$a + b = 16$

The values of (a, b), such that a and b are co-prime are (1, 15), (3, 13), (5, 11) and (7, 9) i.e., 4 pairs.

Ans : (4)

Directions for questions 12 to 15: Select the correct alternative from the given choices.

If the breadth of a rectangle is increased by 5 cm, keeping its length constant, its area increases by 25 sq.cm. If the length is increased by 5 cm, keeping its breadth constant, its area increases by 20 sq.cm. Find the area of the rectangle (in sq.cm).

- a) 20
- b) 25
- c) 30
- d) 35

Solution:

Let the length and breadth of the rectangle be l cm and b cm respectively.
$l(b + 5) = lb + 25$
$l = 5$
$b(l + 5) = lb + 20$
$b = 4$
The area of the rectangle = $lb = 20$ sq.cm

Choice (A)

Directions for questions 12 to 15: Select the correct alternative from the given choices.

How many fractions of the form

$\frac{a}{b}$

(where a and b are single digit natural numbers) can be formed such that no two fractions have the same value and the value of each fraction is less than 2?

- a) 26
- b) 28
- c) 38
- d) 41

Solution:

The numerator, the possible denominators and the number of possible values are tabulated below:

Number	Denominator	Number of possible values
1	1,2,3,4,5,6,7,8,9	9
2	3,5,7,9	4
3	2,4,5,7,8	5
4	3,5,7,9	4
5	3,4,6,7,8,9	6
6	5,7	2
7	4,5,6,8,9	5
8	5,7,9	3
9	5, 7, 8	3
	total	41

Thus there are 41 such fractions. Choice (D)

Directions for questions 12 to 15: Select the correct alternative from the given choices.

In an attempt to optimise the cost of production, an employer reduces the number of employees and increases the salaries of the remaining employees. The old and new number of employees are in the ratio 7 :4. The old and the new salaries are in the ratio 16 : 21. Does the salary bill increase or decrease and by how much?

- a) $33^1/_2$%, decrease
- b) 25%, decrease
- c) 25%, increase
- d) $33^1/_3$%, increase

Solution:

Salary bill = (Number of employees) × (Average sales).
Ratio of salaries paid before the changes and after the changes = $(7 \times 16) : (4 \times 21) = 112 : 84$.
Hence, the salary bill decreases. It decreases by a percentage which is $\left(\frac{112-84}{112}\right) \times 100 = 25\%$ Choice (B)

Directions for questions 12 to 15: Select the correct alternative from the given choices.

There are two identical escalators, one of which is ascending and the other is descending. A person took 96 seconds to go down the ascending escalator and 24 seconds to go down the descending escalator, both times walking at the same speed. If the speed of each escalator is the same, then the time taken to go up / down either of the escalators when they are switched off is _____ sec

- a) 50.4
- b) 38.4
- c) 76.8
- d) 58.8

Solution:

Let the speed of the person be x steps/sec
Let the speed of each escalator be e steps/sec
The escalator are identical
∴ each escalator has the same number of steps. Let us say each has S steps.
The person took 96 seconds to go down the ascending escalator. As he travelled against the escalator, his speed would be (x – e) steps/sec.
The person took 24seconds to go down the descending escalator. As he travelled along the escalator his speed would be (x + e) steps/sec.
$S = 96(x - e) = 24(x + e)$

$x = \frac{5}{3}e$

When the escalators are switched off, the time taken to go up/down using either of the

$$\text{escalators} = \frac{s}{x} = \frac{24(x+e)}{x} = \frac{\frac{192}{3}e}{\frac{5}{3}e} = 38.4\text{sec}$$

Choice (B)